



ENME601 Thermodynamics

Week 5 - Mass and Energy Analysis of Control Volumes

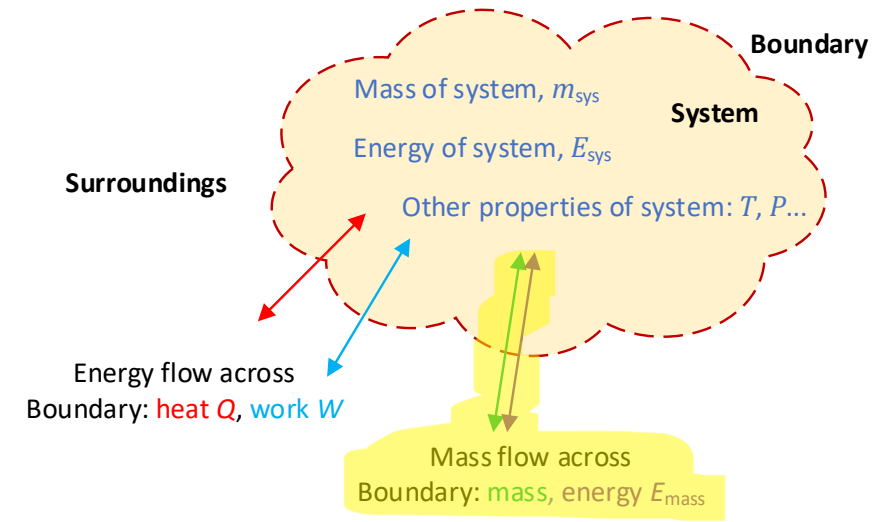
Presenter: Dr. Jay Wang

Objectives

- Develop the **conservation of mass principle** and apply it to steady flow control volumes.
- Apply the first law of thermodynamics as the statement of the **conservation of energy principle to steady flow control volumes**.
- Identify the **energy carried by a fluid stream** crossing a control surface as the sum of **internal energy**, **flow work**, **kinetic energy**, and **potential energy** of the fluid and to relate the combination of the internal energy and the flow work to the property enthalpy.
- Solve energy balance problems for **common steady-flow devices** such as **nozzles**, **compressors**, **turbines**, **throttling valves**, **mixing chambers**, **heat exchangers** and **pipe & duct flow**.

Introduction

- In the topic 4, we considered the first law energy balance applied to **closed systems** (no mass flow across boundary) and we considered the specific heat capacity and its relationship to internal energy and enthalpy for ideal gases, liquids and solids.
- In this topic we consider **open systems that include mass flow across the system boundary**
- We consider **steady flow systems** as a simplification of a more general open system.
- We develop and apply the **conservation of mass principle** and **conservation of energy principle** to **steady flow systems**.



Energy can cross the boundary in two forms: heat Q (red arrow) and work W (blue arrow). These arrows indicate that the surroundings can either add or remove heat and do work on the system or vice versa.

There is also mass flow across the boundary. Mass entering or leaving the system carries its own energy E_{mass} (e.g. internal energy, kinetic energy).

Steady Flow Systems/Processes

- A large number of engineering devices (turbines, compressors, nozzles etc) operate for long periods of time under *steady conditions* i.e. **no changes with respect to time**. The *steady flow process* can model these devices.
- A steady flow process is a process where a *fluid flows through a control volume steadily*.
- Characteristics of steady flow system
 - **No properties** within the control volume **change with time**. (They may vary from point to point).
 - **Volume, mass and total energy content of the system (control volume) remain constant**.
 - **Properties** of mass flows in and out of the system remain **constant at the boundaries**.
 - **Heat and work interactions (across boundary) do not change with time**.



A rotary flow-device (a steam turbine/generator), can be treated as a steady-flow control volume. In a turbine, steam or gas enters at a high pressure and temperature and leaves at a lower pressure and temperature while rotating blades attached to a shaft.



A compressor, which is a steady-flow device.

Steady Flow Process/System

Open-system

Steady = no change
with respect to time

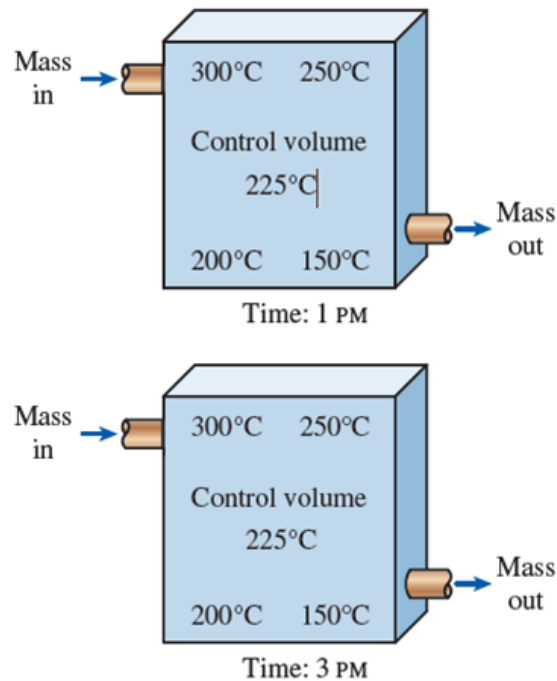


FIGURE 1-32

During a steady-flow process, fluid properties within the control volume may change with position but not with time.

Steady flow:
no change in mass or
energy inside system

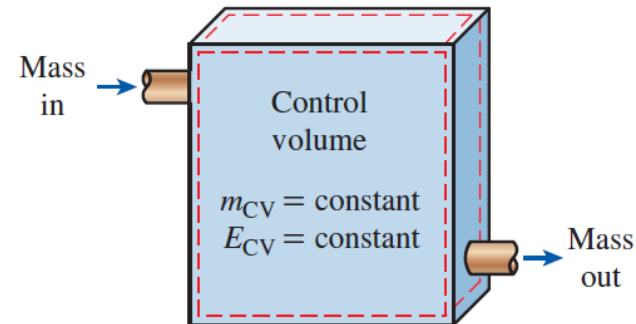


FIGURE 5-18

Under steady-flow conditions, the mass and energy contents of a control volume remain constant.

Steady flow:
Fluid properties at
inlets and exits do not
change with time

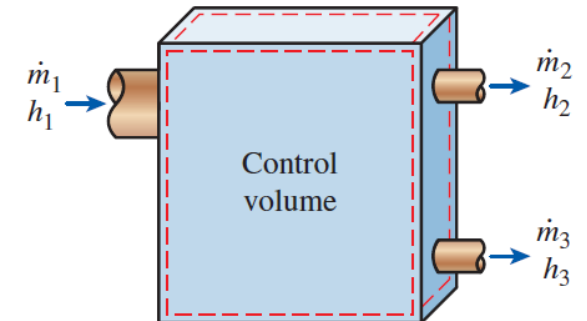


FIGURE 5-19

Under steady-flow conditions, the fluid properties at an inlet or exit remain constant (do not change with time).

Conservation of Mass Principle

Mass and Volume flow rates

Mass flow rate	Volume flow rate
$\dot{m} = \rho v_{\text{avg}} A \quad (\text{kg/s})$ $\dot{m} = \rho \dot{V} = \frac{\dot{V}}{v} \quad (\text{kg/s})$	$\dot{V} = v_{\text{avg}} A \quad (\text{m}^3/\text{s})$ Where: ρ = density of the fluid (kg/m^3) v_{avg} = average or bulk velocity in pipe/duct (m/s) A = cross-sectional area of pipe/duct (m^2) v = specific volume of fluid (m^3/kg)

Fluid flow:

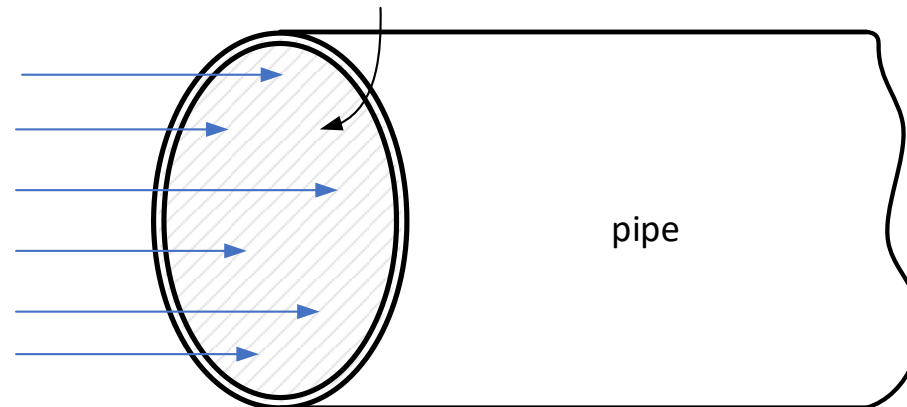
density, ρ

velocity, v

mass flow rate, $\dot{m}$

volume flowrate, $\dot{V}$

cross sectional area, A



Conservation of Mass Principle

Mass like energy is a conserved property

- Generally, for an open system (and for our model of a thermodynamic system) the conservation of mass principle can be expressed as:

$$\left(\begin{array}{c} \text{total mass entering} \\ \text{system in } \Delta t \end{array} \right) - \left(\begin{array}{c} \text{total mass leaving} \\ \text{system in } \Delta t \end{array} \right) = \left(\begin{array}{c} \text{net change in mass} \\ \text{in system in } \Delta t \end{array} \right)$$

Or
$$\sum m_{\text{in}} - \sum m_{\text{out}} = \Delta m_{\text{system}}$$

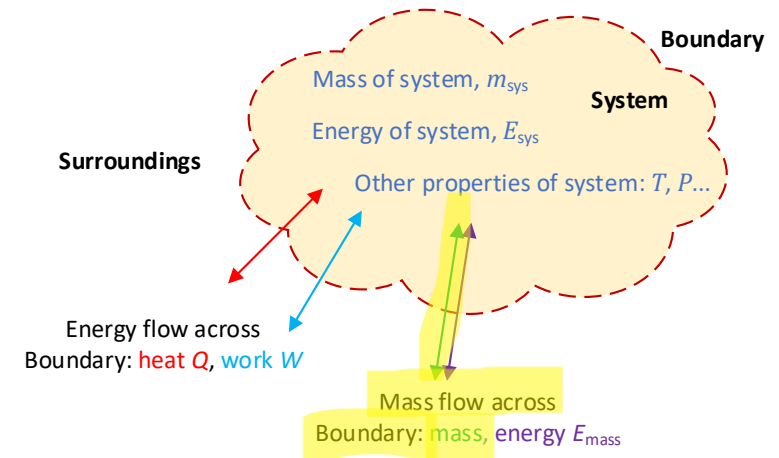
- In rate form this becomes:

$$\left(\begin{array}{c} \text{total mass flow} \\ \text{entering system} \end{array} \right) - \left(\begin{array}{c} \text{total mass flow} \\ \text{leaving system} \end{array} \right) = \left(\begin{array}{c} \text{rate of change of} \\ \text{mass in system} \end{array} \right)$$

Or
$$\sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}} = dm_{\text{system}}/dt$$

- For a **steady flow process** the mass in the control volume is constant so $dm_{\text{system}}/dt = 0$ and so

$$\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$$



Mass Balance for Steady Flow Systems/Processes

- For a **steady flow system/process** the total amount of **mass contained in the control volume remains constant**, so total mass flow **into** control volume **is equal to** total mass flow **out** of control volume:

$$\sum \dot{m}_i = \sum \dot{m}_e$$

Where subscript i = inlets and e = exits or outlets

- In many cases there is just one inlet (1) and one outlet (2) so

$$\dot{m}_1 = \dot{m}_2 ,$$

$$\rho_1 v_1 A_1 = \rho_2 v_2 A_2 ,$$

Volumetric
flow rate

$$\frac{v_1 A_1}{v_1} = \frac{v_2 A_2}{v_2}$$

Note:

v = velocity (m/s)

v = specific volume (m^3/kg) ($= 1/\rho$)

$$\rho = \frac{1}{v}$$

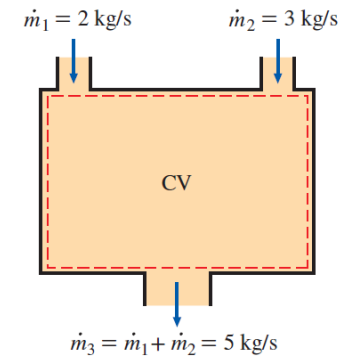


FIGURE 5–8

Conservation of mass principle for a two-inlet–one-outlet steady-flow system.

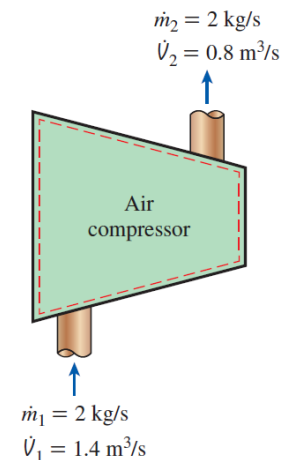


FIGURE 5–9

During a steady-flow process, volume flow rates are not necessarily conserved, although mass flow rates are.

Flow Work and the Energy of a Flowing Fluid

- **Flow work** is the work associated with pushing mass **into** or **out of** the control volume.
- Consider a fluid element of volume V and cross-sectional area, A . Fluid upstream at pressure P is pushing it into the control volume. (And force $F = PA$)
- The work done in pushing the volume element a distance L into the control volume is therefore:

$$W_{\text{flow}} = FL = PAL = PV \quad (\text{kJ})$$

On a per unit mass: $w_{\text{flow}} = Pv \quad (\text{kJ/kg})$

Where v is the specific volume of the fluid

- Note: this is equivalent to the **flow energy**, that we discussed as a **component of the mechanical energy of a flowing fluid**.

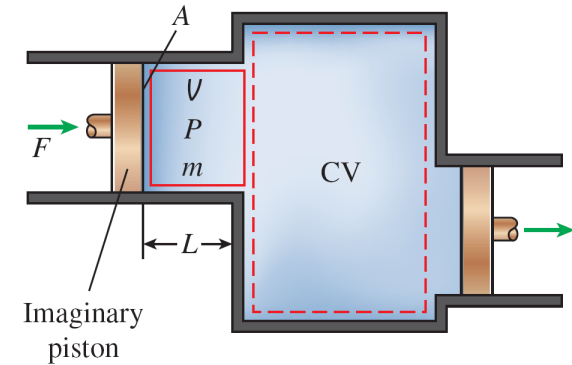
If mass is flowing with a mass flow rate $\dot{m}$ (kg/s), the rate of flow energy is:

$$\dot{E}_{\text{flow}} = \dot{m} \cdot Pv$$

- Units: kJ/s = kW

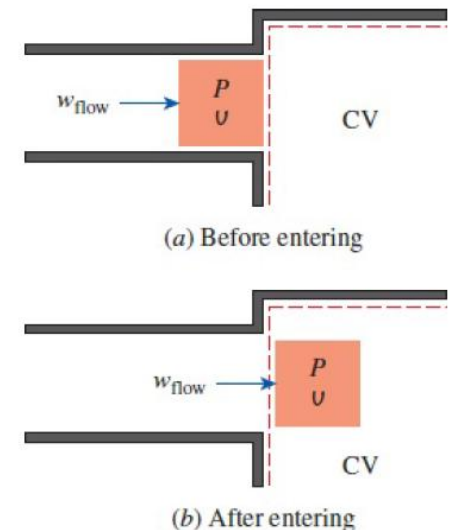
How mass enters the control volume?

An imaginary piston is introduced at the control surface boundary.



1. Before entering, a fluid element (small control mass) is just outside the CV boundary

2. Once it enters the CV, it brings energy associated with mass transport into the CV.



Flow Work and the Energy of a Flowing Fluid

- If we consider the small volume element in the previous slide as **not flowing** i.e. as a **normal system** with **internal**, **kinetic** and **potential** energies, then:

closed system

$$e_{\text{non-flow}} = u + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

- To cause the volume element **to flow (in a pipe)** then we need to include the **flow energy Pv** to give the total energy of the flowing fluid:

$$e_{\text{flow}} = Pv + u + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

And since $h = u + Pv$

$$e_{\text{flow}} = h + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

- So, the **total energy flow rate, transported the mass** is

$$\dot{E}_{\text{mass}} = \dot{m}e_{\text{flow}} = \dot{m}(h + e_K + e_P) = \dot{m} \left(h + \frac{v^2}{2} + gz \right) \quad (\text{kW})$$

Flow Work and the Energy of a Flowing Fluid

- Alternatively, the total specific mechanical energy of a flowing fluid was made up of flow energy, kinetic energy and potential energy

$$e_{\text{mech}} = Pv + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

- If we now include the thermal energy component (u) of the fluid, then the total energy of a flowing fluid is:

$$e_{\text{flow}} = u + Pv + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

And since $h = u + Pv$

$$e_{\text{flow}} = h + \frac{v^2}{2} + gz \quad (\text{kJ/kg})$$

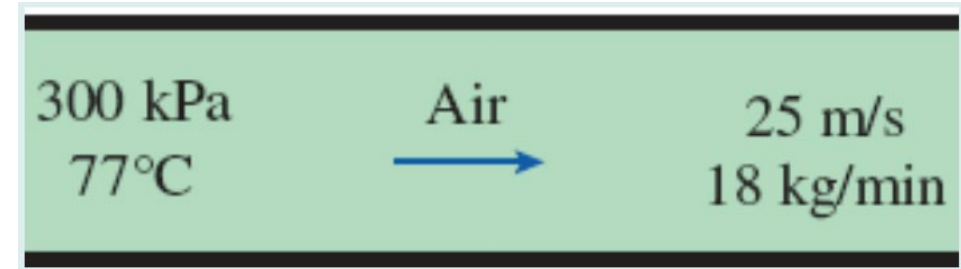
- So, the total energy flow rate, transported the mass is

$$\dot{E}_{\text{mass}} = \dot{m}e_{\text{flow}} = \dot{m}(h + e_K + e_P) = \dot{m} \left(h + \frac{v^2}{2} + gz \right) \quad (\text{kW})$$

Short practice 1:

Hints:

Air (ideal gas) flows steadily in a pipe at 300 kPa, 77°C, and 25 m/s at a rate of 18 kg/min. Determine (a) the diameter of the pipe, (b) the rate of flow energy (in kW), (c) the rate of energy transport by mass (in kW)



Assumptions: **1** The flow is steady. **2** The potential energy is negligible.

Properties: The properties of air are $R = 0.287 \text{ kJ/kg}\cdot\text{K}$ and $C_p = 1.008 \text{ kJ/kg}\cdot\text{K}$ (at 350 K from Table A-2b).

Solutions:

(a) The diameter is determined as follows:

$$v = \frac{RT}{P} = \frac{(0.287 \text{ kJ/kg}\cdot\text{K})(77 + 273 \text{ K})}{300 \text{ kPa}} = 0.3349 \text{ m}^3/\text{kg}$$

$$A = \frac{\dot{m}v}{V} = \frac{(18/60 \text{ kg/s})(0.3349 \text{ m}^3/\text{kg})}{25 \text{ m/s}} = 0.004018 \text{ m}^2$$

$$D = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4(0.004018 \text{ m}^2)}{\pi}} = \mathbf{0.0715 \text{ m}}$$

Solutions:

(b) The rate of flow energy is determined from

$$\dot{W}_{\text{flow}} = \dot{m}Pv = (18/60 \text{ kg/s})(300 \text{ kPa})(0.3349 \text{ m}^3/\text{kg}) = \mathbf{30.14 \text{ kW}}$$

(c) The rate of energy transport by mass is

$$\begin{aligned}\dot{E}_{\text{mass}} &= \dot{m}(h + \text{ke}) = \dot{m}\left(c_p T + \frac{1}{2}V^2\right) \\ &= (18/60 \text{ kg/s})\left[(1.008 \text{ kJ/kg}\cdot\text{K})(77 + 273 \text{ K}) + \frac{1}{2}(25 \text{ m/s})^2\left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2}\right)\right] \\ &= \mathbf{105.94 \text{ kW}}\end{aligned}$$

h can also be found from Table A17.

Energy Analysis of Steady Flow Systems

- For a **steady flow process** the total energy of the control volume, remains constant i.e. the rate of change of energy of the control volume (system) = 0. Writing the energy balance equation in rate form:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{rate of net energy transfer by heat, work and mass}} = \underbrace{\cancel{\frac{dE_{\text{system}}}{dt}}^0_{\text{rate of change in internal kinetic, potential, etc energies}}$$

open system

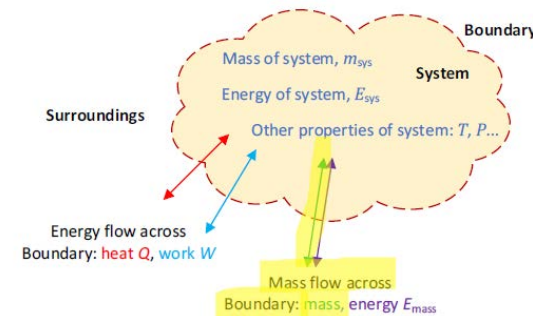
- Thus, the rate of energy entering the control volume in all forms (heat, work and mass transfer) must be equal to the rate of energy leaving it

$$\underbrace{\dot{E}_{\text{in}}}_{\text{rate of net energy transfer by heat, work and mass}} = \underbrace{\dot{E}_{\text{out}}}_{\text{rate of net energy transfer by heat, work and mass}}$$

This can be written as:

$$\dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \sum \dot{E}_{\text{mass, in}} = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \sum \dot{E}_{\text{mass, out}}$$

$$\Rightarrow \dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \sum \dot{m}_{\text{in}} e_{\text{flow, in}} = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \sum \dot{m}_{\text{out}} e_{\text{flow, out}}$$



Energy Analysis of Steady Flow Systems

$$\dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \sum \dot{m}_{\text{in}} e_{\text{flow,in}} = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \sum \dot{m}_{\text{out}} e_{\text{flow,out}}$$

- Note: e_{flow} is the total energy of a **flowing** fluid so $\dot{m}e_{\text{flow}} (= \dot{E}_{\text{mass}})$ is the energy transported with mass flow to/from the control volume per unit time, So

$$\dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}} = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}}$$

- If we use the formal sign convention (**heat transfer to a system and work done by a system are positive and heat transfer from a system and work done on a system are negative**). And rearrange, then we can write this as:

$$\dot{Q} - \dot{W} = \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}} - \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}}$$

Where:

- $\dot{Q}$ is net heat input: $\dot{Q}_{\text{in}} - \dot{Q}_{\text{out}}$
- $\dot{W}$ is net work output: $\dot{W}_{\text{out}} - \dot{W}_{\text{in}}$

Energy Analysis of Steady Flow Systems

$$\dot{Q} - \dot{W} = \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}} - \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}}$$

General
form of
SFEE

- For a **single stream** (one inlet, one outlet), steady flow process then subscript 1 represents the inlet and subscript 2 represents the outlet and:

$$\dot{Q} - \dot{W} = \dot{m} \left[h_2 - h_1 + \frac{v_2^2 - v_1^2}{2} + g(z_2 - z_1) \right] \quad (\text{kW})$$

or more simply:

$$\dot{Q} - \dot{W} = \dot{m} [\Delta h + \Delta e_K + \Delta e_P] \quad (\text{kW})$$

SFEE: one inlet,
one outlet

In some cases we can ignore the changes in kinetic and potential energy so:

$$\dot{Q} - \dot{W} = \dot{m} \Delta h \quad (\text{kW})$$

- These steady flow energy equations are often referred to as SFEE's



Application of the steady flow energy equation (SFEE) to various engineering devices

Steady Flow Engineering Devices

Nozzles and Diffusers

- Nozzle = a device that **increases the velocity of a fluid** at the expense of pressure (**decrease the pressure**).
- Diffuser = a device that **decreases the velocity of a fluid** and **increases its pressure**.
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$\dot{Q} \cong 0$	velocities are high, so high speed process and heat does not have time to transfer, so essentially adiabatic
$\dot{W} = 0$	no forms of work cross the boundary (shaft, electrical etc)
$\Delta e_P \cong 0$	usually minor changes in elevation for a nozzle
$\Delta e_K > 0$	velocity changes are significant so we can't ignore kinetic energy changes

- SFEE for nozzles and diffusers

$$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2} \quad \text{----->} \quad (h_2 - h_1) = - \left(\frac{v_2^2 - v_1^2}{2} \right)$$

Nozzles and Diffusers

negative

positive

In a nozzle $\rightarrow h_2 - h_1 < 0$ (enthalpy drop, velocity gain).

In a diffuser $\rightarrow h_2 - h_1 > 0$ (enthalpy rise, velocity drop).

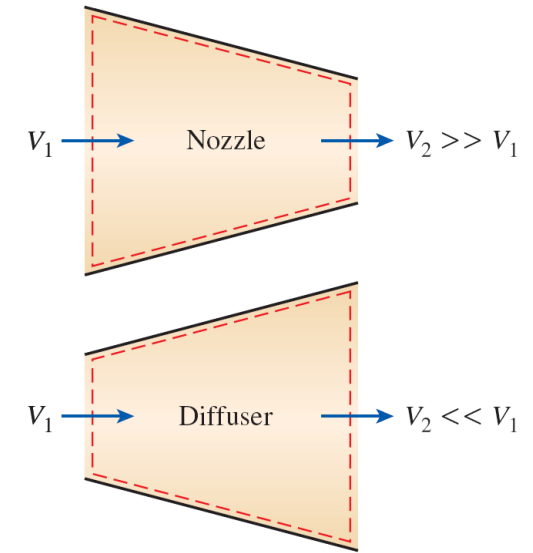


Figure 6–26: Nozzles and diffusers are shaped so that they cause large changes in fluid velocities and thus kinetic energies.



Jet engine diffuser

Short practice 2:

Air (ideal gas) at 10°C and 80 kPa enters the diffuser of a jet engine steadily with a velocity of 200 m/s . The inlet area of the diffuser is 0.4 m^2 . The air leaves the diffuser with a velocity that is very small compared with the inlet velocity. Determine (a) the mass flow rate of the air and (b) the temperature of the air leaving the diffuser.

- Assumptions:

1. This is a steady-flow process since there is no change with time at any point.
2. The potential energy change is zero.
3. Heat transfer is negligible.
4. Kinetic energy at the diffuser exit is negligible.
5. There are no work interactions.

Solutions:

(a) To determine the mass flow rate, we need to find the specific volume of the air first. This is determined from the ideal-gas relation at the inlet conditions:

$$v_1 = \frac{RT_1}{P_1} = \frac{(0.287 \text{ kPa} \cdot \text{m}^3/\text{kg} \cdot \text{K})(283 \text{ K})}{80 \text{ kPa}} = 1.015 \text{ m}^3/\text{kg}$$

Then,

$$\dot{m} = \frac{1}{v_1} V_1 A_1 = \frac{1}{1.015 \text{ m}^3/\text{kg}} (200 \text{ m/s})(0.4 \text{ m}^2) = \mathbf{78.8 \text{ kg/s}}$$

Since the flow is steady, the mass flow rate through the entire diffuser remains constant at this value.

Solutions:

(b) Under stated assumptions and observations, the energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\frac{dE_{\text{system}}}{dt}}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} \overset{0 \text{ (steady)}}{=} 0$$
$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$
$$\dot{m} \left(h_1 + \frac{V_1^2}{2} \right) = \dot{m} \left(h_2 + \frac{V_2^2}{2} \right) \quad (\text{since } \dot{Q} \cong 0, \dot{W} = 0, \text{ and } \Delta pe \cong 0)$$
$$h_2 = h_1 - \frac{V_2^2 - V_1^2}{2}$$

The exit velocity of a diffuser is usually small compared with the inlet velocity ($V_2 \ll V_1$); thus, the kinetic energy at the exit can be neglected. The enthalpy of air at the diffuser inlet is determined from the air table (Table A-17) to be

$$h_1 = h_{@ 283 \text{ K}} = 283.14 \text{ kJ/kg}$$

Solutions:

Substituting, we get

$$\begin{aligned}h_2 &= 283.14 \text{ kJ/kg} - \frac{0 - (200 \text{ m/s})^2}{2} \left(\frac{1 \text{ kJ/kg}}{1000 \text{ m}^2/\text{s}^2} \right) \\&= 303.14 \text{ kJ/kg}\end{aligned}$$

From Table A–17, the temperature corresponding to this enthalpy value is

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$$T_2 = \mathbf{303 \text{ K}}$$

Discussion This result shows that the temperature of the air increases by about 20°C as it is slowed down in the diffuser. The temperature rise of the air is mainly due to the conversion of kinetic energy to internal energy.

Steady Flow Engineering Devices

Turbines and Compressors

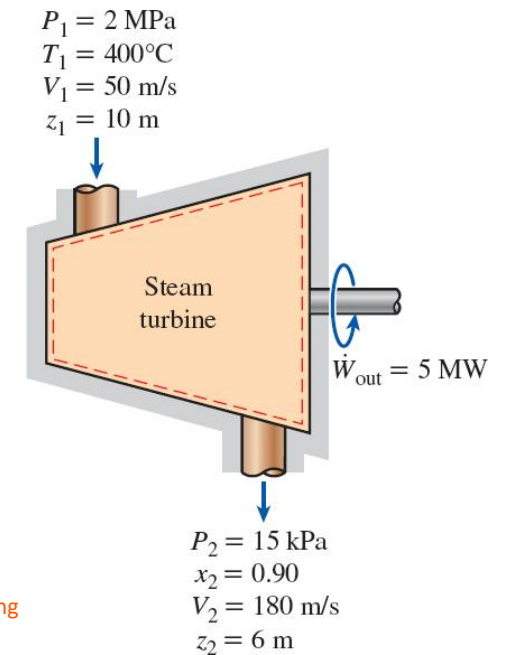
- Turbine = **produces work** (power) to drive generators, machinery, marine power. Can be steam or gas.
- Compressor = used to **increase the pressure of a gas**
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$\dot{Q} \cong 0$	Usually small relative to work produced or consumed. If intentional cooling in a compressor then $\dot{Q} \neq 0$. (like spraying water, intercooler, or cooling jackets)
$\dot{W} > 0$	Turbines produce work. Compressors consume work.
$\Delta e_P \cong 0$	usually minor changes in elevation for a turbine or compressor
$\Delta e_K \cong 0$	velocities can be high in turbines (much lower in compressors) but Δe_K small relative to Δh .

- SFEE for turbines and compressors (simplest form)

$$\dot{W} = -\dot{m}\Delta h \quad \text{Turbines and Compressors}$$

$$-\dot{W}_s \approx \dot{m} \Delta h \quad \Rightarrow \quad \dot{W}_s \approx \dot{m} (h_1 - h_2)$$

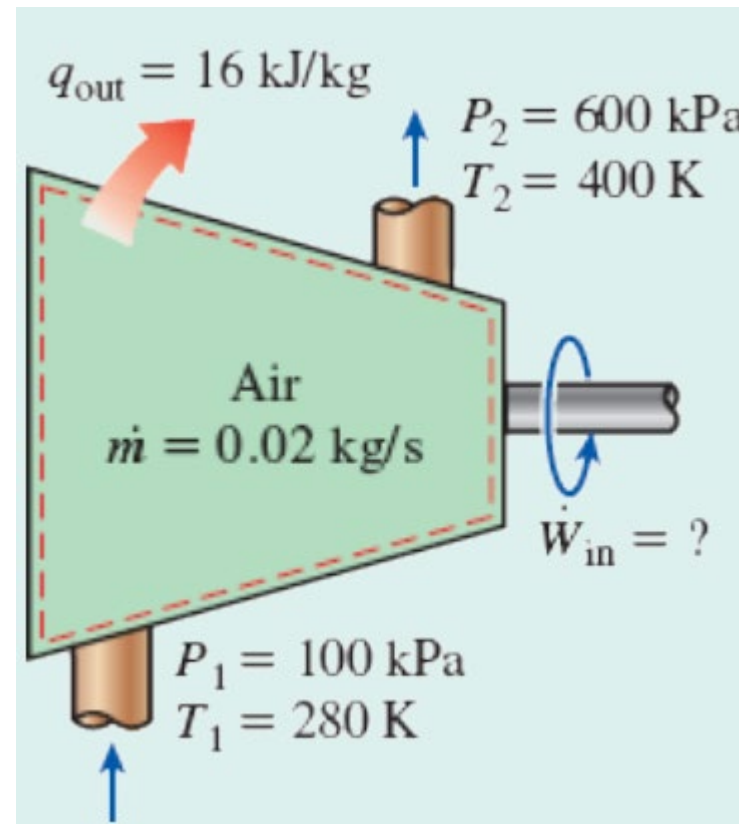


Turbine blades

Short practice 3:

Air (ideal gas) at 100 kPa and 280 K is compressed steadily to 600 kPa and 400 K. The mass flow rate of the air is 0.02 kg/s, and a heat loss of 16 kJ/kg occurs during the process. Assuming the changes in kinetic and potential energies are negligible, determine the necessary power input to the compressor.

Hint:



- Assumptions:

- 1.** This is a steady-flow process since there is no change with time at any point.
- 2.** The kinetic and potential energy change is zero.

Solutions:

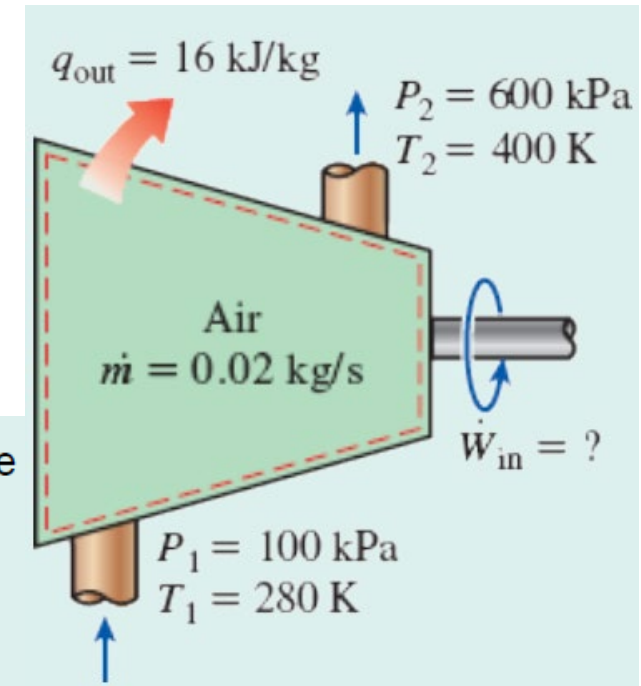
Under stated assumptions and observations, the energy balance for this steady-flow system can be expressed in the rate form as

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\frac{dE_{\text{system}}}{dt}}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} \xrightarrow{0 \text{ (steady)}} = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{W}_{\text{in}} + \dot{m}h_1 = \dot{Q}_{\text{out}} + \dot{m}h_2 \quad (\text{since } \Delta \text{ke} = \Delta \text{pe} \cong 0)$$

$$\dot{W}_{\text{in}} = \dot{m}q_{\text{out}} + \dot{m}(h_2 - h_1)$$



Solutions:

The enthalpy of an ideal gas depends on temperature only, and the enthalpies of the air at the specified temperatures are determined from the air table (Table A-17) to be

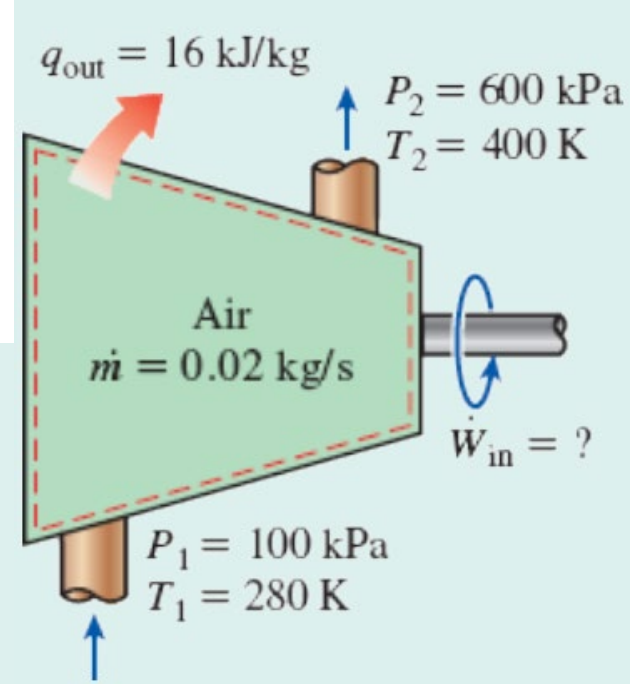
$$h_1 = h_{@ 280 \text{ K}} = 280.13 \text{ kJ/kg}$$

$$h_2 = h_{@ 400 \text{ K}} = 400.98 \text{ kJ/kg}$$

Substituting, the power input to the compressor is determined to be

$$\begin{aligned}\dot{W}_{\text{in}} &= (0.02 \text{ kg/s})(16 \text{ kJ/kg}) + (0.02 \text{ kg/s})(400.98 - 280.13) \text{ kJ/kg} \\ &= \mathbf{2.74 \text{ kW}}\end{aligned}$$

Discussion Note that the mechanical energy input to the compressor manifests itself as a rise in enthalpy of air and heat loss from the compressor.



Steady Flow Engineering Devices

Throttling valves

- These are flow restricting devices used principally to **reduce the pressure** in a flow stream. E.g. valve, porous plug, capillary tube
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$\dot{Q} \cong 0$	Small devices, with high velocities so essentially adiabatic
$\dot{W} = 0$	No work involved
$\Delta e_P \cong 0$	usually minor changes in elevation for a throttling valve
$\Delta e_K \cong 0$	Change in velocity can be high in throttles but Δe_K small relative to Δh .

- SFEE for throttles

$$0 = \dot{m}\Delta h \Rightarrow \Delta h = 0$$

so

$$h_1 = h_2 \quad \text{Throttle}$$

- Throttles are constant enthalpy or **isenthalpic** devices

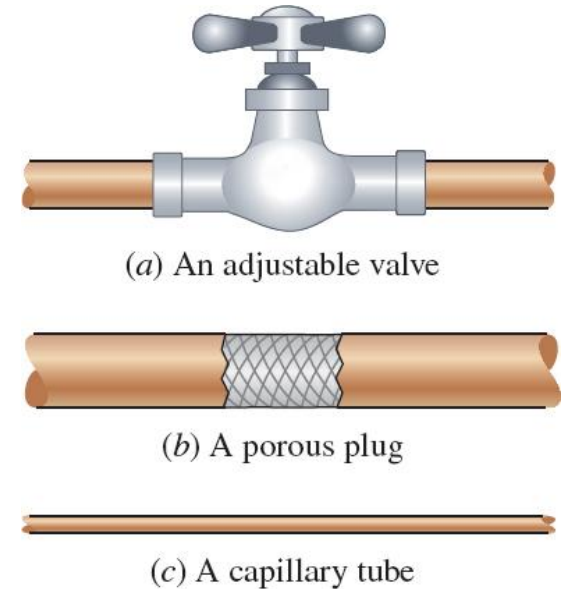


Figure 6–32: Throttling valves are devices that cause large pressure drops in the fluid.

Same enthalpy
Significant pressure drop

Steady Flow Engineering Devices

Mixing Chambers

- The mixing of two or more streams of fluid at different temperatures, specifically to achieve an intermediate temperature.
- SFEE, multiple inlets & single outlet (general form):

$$\dot{Q} - \dot{W} = \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}} - \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}}$$

$\dot{Q} \cong 0$	Often insulated and high velocities so essentially adiabatic
$\dot{W} = 0$	No work involved
$\Delta e_p \cong 0$	usually minor changes in elevation for a mixing chamber
$\Delta e_K \cong 0$	Some changes in velocity but Δe_K small relative to Δh .

- SFEE for a mixing chamber

$$\underbrace{\sum \dot{m}_{\text{in}} h_{\text{in}}}_{\text{for each inlet}} = \underbrace{\sum \dot{m}_{\text{out}} h_{\text{out}}}_{\text{for each outlet}}$$

Mixing
Chamber

And also apply a mass balance
 $\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$

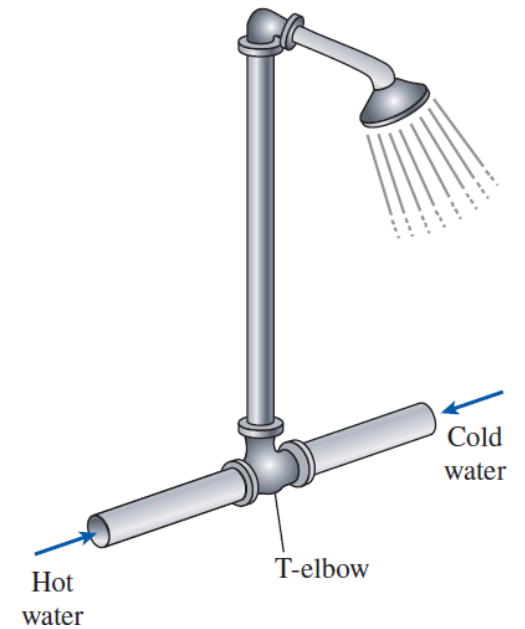


Figure 6–35: The T-elbow of an ordinary shower serves as the mixing chamber for the hot- and the cold-water streams.

Steady Flow Engineering Devices

Heat Exchangers

- A **hotter fluid exchanges heat with a cooler fluid**. The fluids are kept separate by a wall or pipe etc.
- SFEE, single inlet & single outlet (consider hot and cold streams separately): $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

$\dot{Q} > 0$	Heat lost by hotter fluid = heat gained by cooler fluid
$\dot{W} = 0$	No work involved
$\Delta e_P \cong 0$	Usually minor changes in elevation for a heat exchanger
$\Delta e_K \cong 0$	Some changes in velocity but Δe_K small relative to Δh .

- SFEE's for a heat exchanger

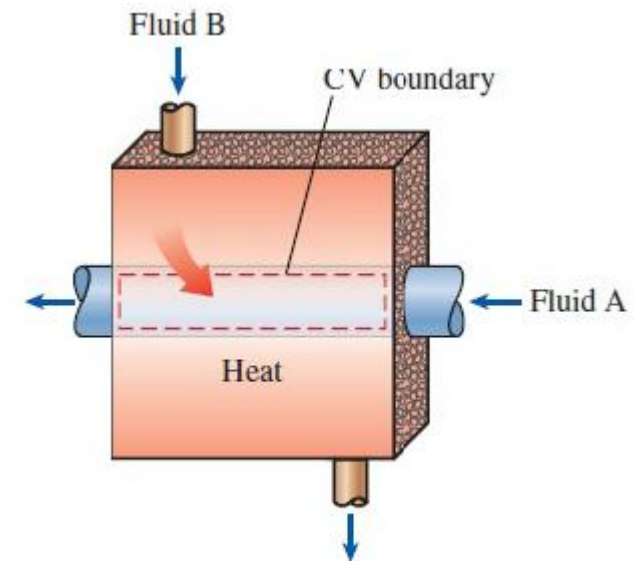
$$\begin{aligned}\dot{Q}_{\text{cold}} &= (\dot{m}\Delta h)_{\text{cold}} = [\dot{m}(h_{\text{out}} - h_{\text{in}})]_{\text{cold}} \\ \dot{Q}_{\text{hot}} &= (\dot{m}\Delta h)_{\text{hot}} = [\dot{m}(h_{\text{out}} - h_{\text{in}})]_{\text{hot}} \\ -\dot{Q}_{\text{hot}} &= \dot{Q}_{\text{cold}}\end{aligned}$$

$$\dot{m}_h(h_{h,\text{in}} - h_{h,\text{out}}) = \dot{m}_c(h_{c,\text{out}} - h_{c,\text{in}})$$

Heat
Exchanger

- **Hot stream:** loses energy → enthalpy decreases
- **Cold stream:** gains energy → enthalpy increases
- Net heat transfer within the CV is **zero** (if insulated):

$$\dot{Q}_{\text{hot}} + \dot{Q}_{\text{cold}} = 0$$



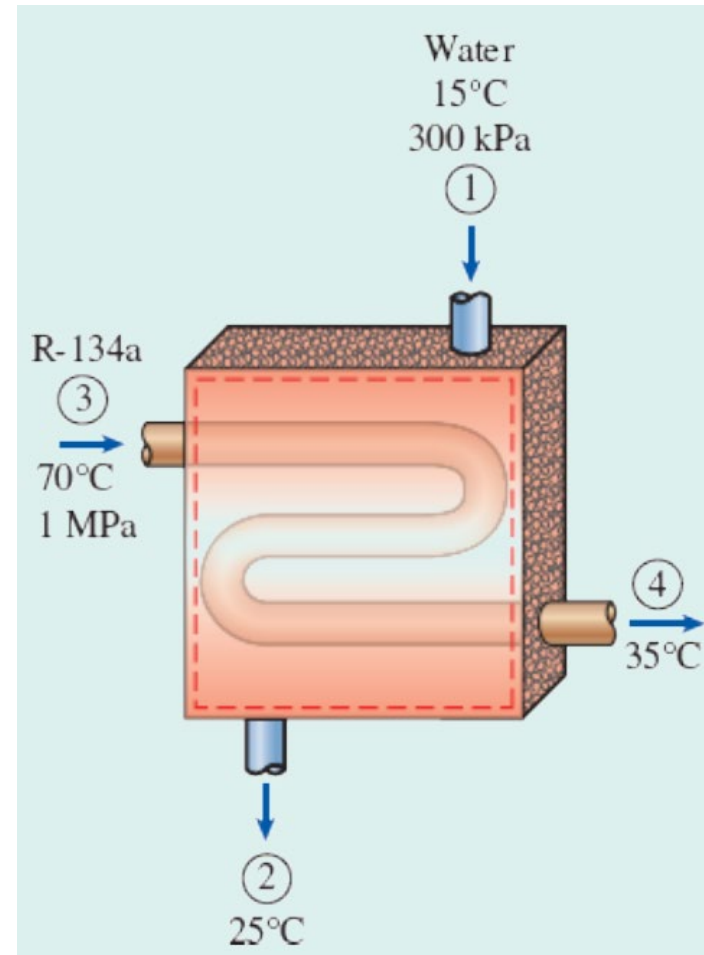
(b) System: Fluid A ($\dot{Q}_{\text{CV}} \neq 0$)

Heat loss by hot fluid =
heat gain by cold fluid

Short practice 4:

Refrigerant-134a is to be cooled by water in a condenser. The refrigerant enters the condenser with a mass flow rate of 6 kg/min at 1 MPa and 70°C and leaves at 35°C. The cooling water enters at 300 kPa and 15°C and leaves at 25°C. Neglecting any pressure drops. To determine:

- (a) the mass flow rate of the cooling water required.
- (b) the heat transfer rate from the refrigerant to water.



- Assumptions:

1. This is a steady-flow process since there is no change with time at any point.
2. The kinetic and potential energy change is zero.
3. Heat losses from the system are negligible.
4. There is no work interaction.

Solutions:

(a) Under the stated assumptions and observations, the mass and energy balances for this steady-flow system can be expressed in the rate form as follows:

Mass balance:

$$\dot{m}_{\text{in}} = \dot{m}_{\text{out}}$$

for each fluid stream since there is no mixing. Thus,

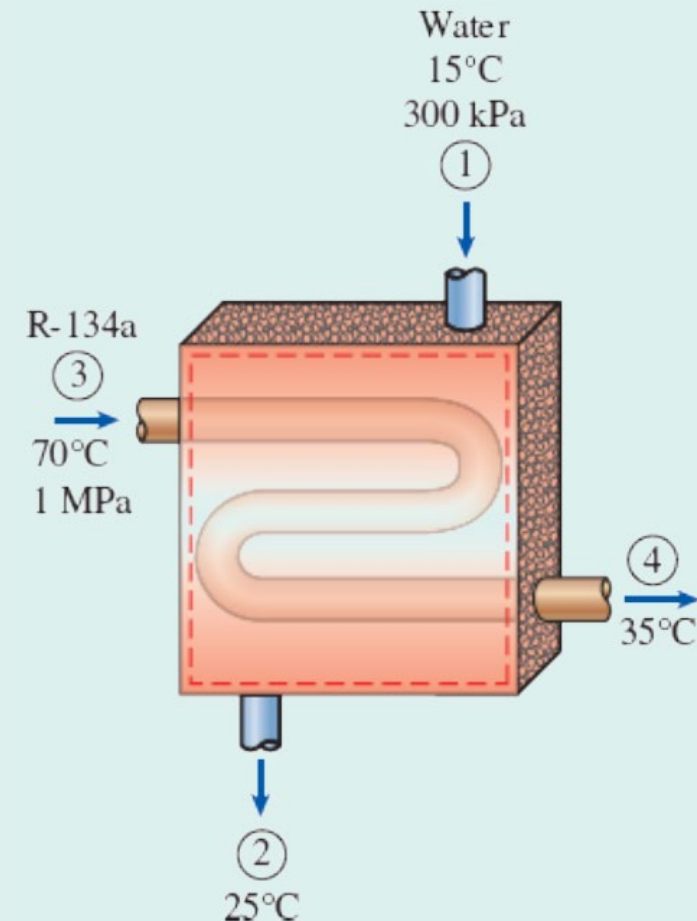
$$\begin{aligned}\dot{m}_1 &= \dot{m}_2 = \dot{m}_w \\ \dot{m}_3 &= \dot{m}_4 = \dot{m}_R\end{aligned}$$

Energy balance:

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{Rate of net energy transfer by heat, work, and mass}} = \underbrace{\frac{dE_{\text{system}}}{dt}}_{\text{Rate of change in internal, kinetic, potential, etc., energies}} \xrightarrow{0 \text{ (steady)}} 0 = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{m}_1 h_1 + \dot{m}_3 h_3 = \dot{m}_2 h_2 + \dot{m}_4 h_4 \quad (\text{since } \dot{Q} \cong 0, \dot{W} = 0, \text{ke} \cong \text{pe} \cong 0)$$



Solutions:

Now we need to determine the enthalpies at all four states. Water exists as a compressed liquid at both the inlet and the exit since the temperatures at both locations are below the saturation temperature of water at 300 kPa (133.52°C). Approximating the compressed liquid as a saturated liquid at the given temperatures, we have

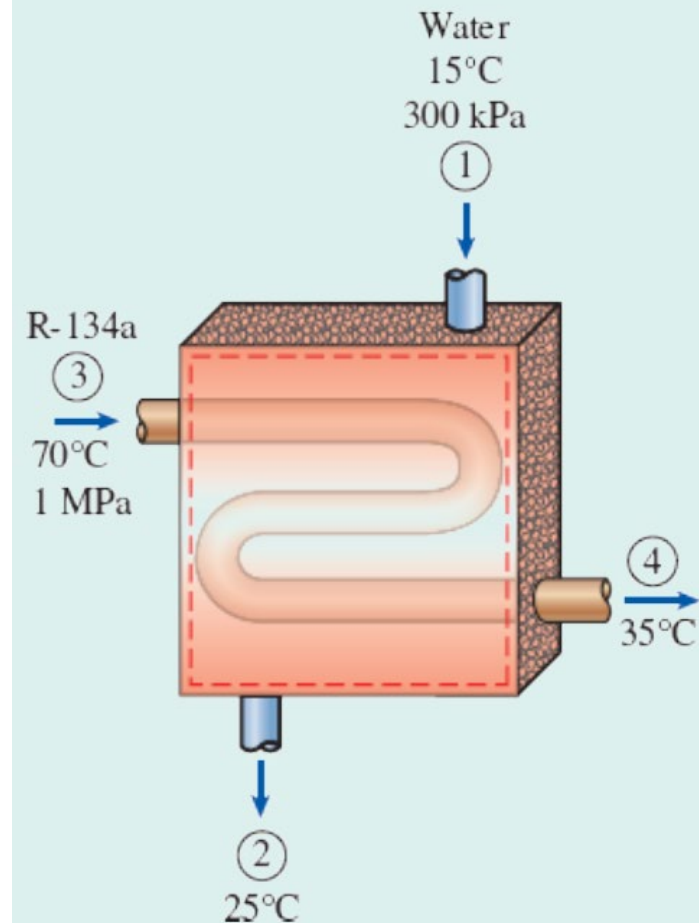
$$\begin{aligned}h_1 &\cong h_{f@15^\circ\text{C}} = 62.982 \text{ kJ/kg} \\h_2 &\cong h_{f@25^\circ\text{C}} = 104.83 \text{ kJ/kg}\end{aligned}\quad \begin{array}{l} \text{(Table} \\ \text{A-4)} \end{array}$$

The refrigerant enters the condenser as a superheated vapor and leaves as a page 237 compressed liquid at 35°C. From refrigerant-134a tables,

$$\left. \begin{array}{l} P_3 = 1 \text{ MPa} \\ T_3 = 70^\circ\text{C} \end{array} \right\} h_3 = 303.87 \text{ kJ/kg} \quad \begin{array}{l} \text{(Table} \\ \text{A-13)} \end{array}$$

$$\left. \begin{array}{l} P_4 = 1 \text{ MPa} \\ T_4 = 70^\circ\text{C} \end{array} \right\} h_4 \cong h_{f@35^\circ\text{C}} = 100.88 \text{ kJ/kg} \quad \begin{array}{l} \text{(Table} \\ \text{A-11)} \end{array}$$

Substituting, we find



Solutions:

$$\dot{m}_w(62.982 - 104.83)\text{kJ/kg} = (6 \text{ kg/min})[(100.88 - 303.87) \text{ kJ/kg}]$$

$$\dot{m}_w = \mathbf{29.1 \text{ kg/min}}$$

(b) To determine the heat transfer from the refrigerant to the water, we have to choose a control volume whose boundary lies on the path of heat transfer. We can choose the volume occupied by either fluid as our control volume. For no particular reason, we choose the volume occupied by the water. All the assumptions stated earlier apply, except that the heat transfer is no longer zero. Then assuming heat to be transferred to water, the energy balance for this single-stream steady-flow system reduces to

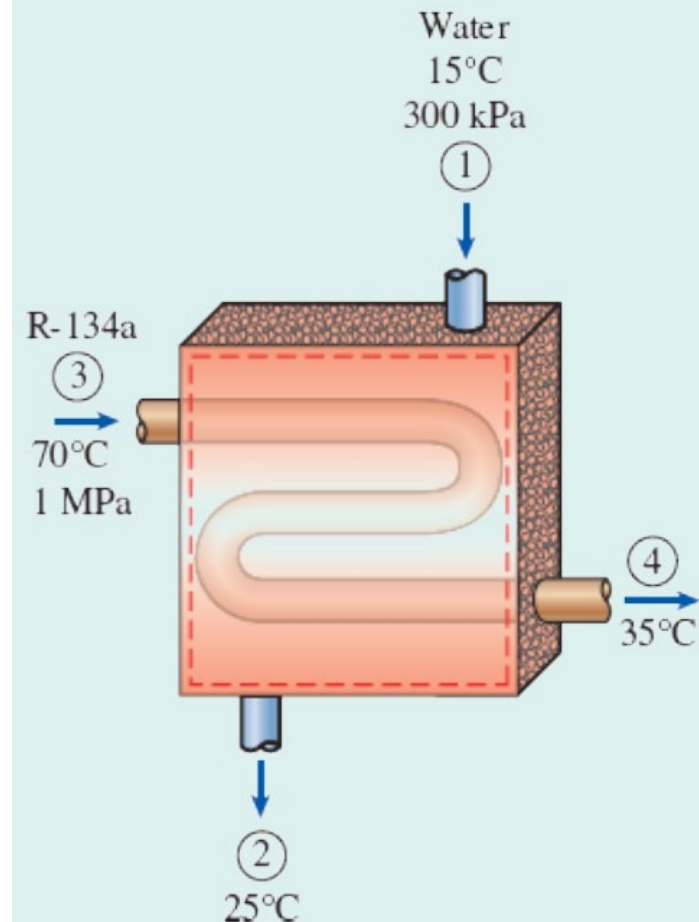
$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\substack{\text{Rate of net energy transfer} \\ \text{by heat, work, and mass}}} = \underbrace{\frac{dE_{\text{system}}}{dt}}_{\substack{\text{Rate of change in internal, kinetic,} \\ \text{potential, etc., energies}}} \xrightarrow{0 \text{ (steady)}} = 0$$

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

$$\dot{Q}_{w,\text{in}} + \dot{m}_w h_1 = \dot{m}_w h_2$$

Rearranging and substituting,

$$\begin{aligned} \dot{Q}_{w,\text{in}} &= \dot{m}_w(h_2 - h_1) = (29.1 \text{ kg/min})[(104.83 - 62.982) \text{ kJ/kg}] \\ &= \mathbf{1218 \text{ kJ/min}} \end{aligned}$$



Steady Flow Engineering Devices

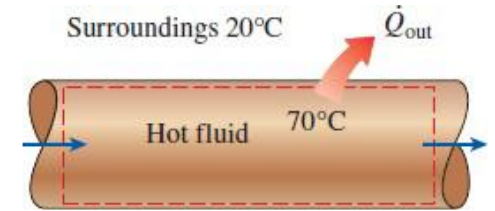
Pipe and Duct flow

- All forms of energy may be involved in the flow of fluids in pipes and ducts, heat transfer, work interactions and changes in kinetic and potential energies
- SFEE, single inlet & single outlet: $\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$

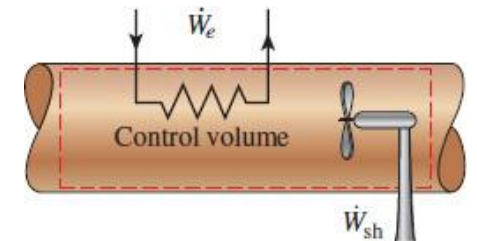
$\dot{Q} > 0$	heat transfer to or from the fluid: boiler tubes, refrigeration coils, duct heaters etc.
$\dot{W} > 0$	if control volume contains a heater ($\dot{W}_e$) or a fan or pump ($\dot{W}_s$) then work interactions do take place.
$\Delta e_P > 0$	pipes and ducts usually undergo elevation changes so potential energy changes could be significant.
$\Delta e_K \cong 0$	if $\dot{Q}$ is significant and there are no large diameter changes then changes in kinetic energy can be insignificant.
- SFEE for pipe and duct flow (no simplification)

$$\dot{Q} - \dot{W} = \dot{m}[\Delta h + \Delta e_K + \Delta e_P]$$

Pipe & Duct
Flow



Heat transfer



Work interactions